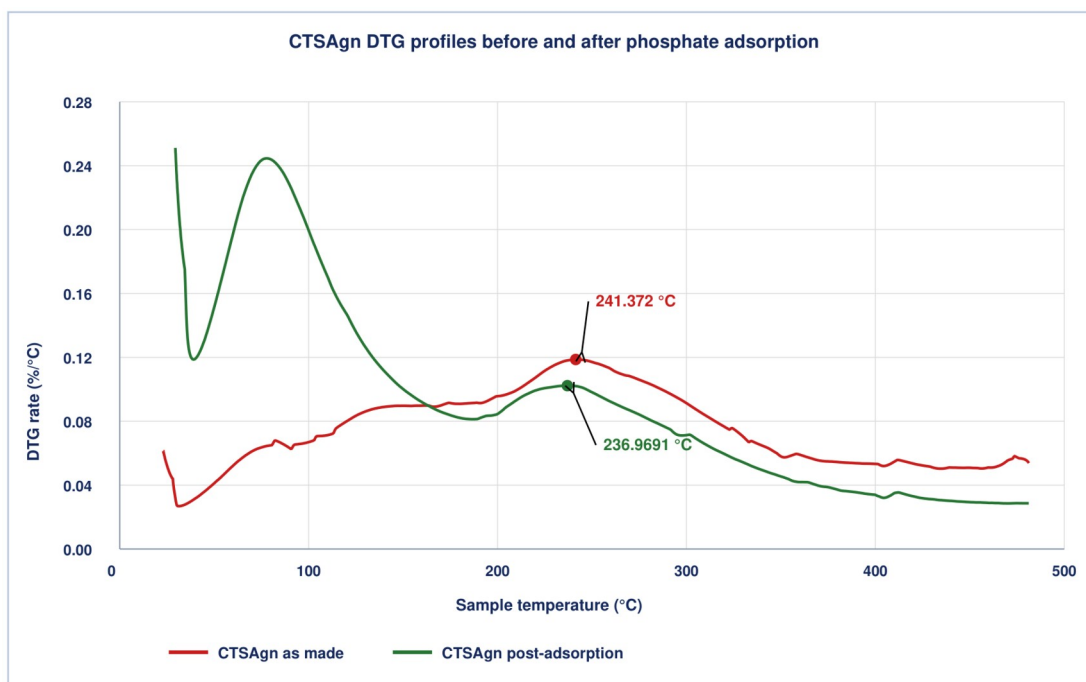


Thermal stability assessment: CTSAgn composite

Primary degradation peaks from the September TGA campaign, read above the water step.

Material	Peak temperature (°C)	Peak DTG rate (%/°C)
Chitosan precursor	305.1665	1.0229
Alginate precursor	246.7935	1.1307
CTSAgn as made	241.3720	0.1187
CTSAgn post-adsorption	236.9691	0.1023



As made, CTSAgn has a peak mass-loss rate 8.6 times lower than chitosan and 9.5 times lower than alginate. After phosphate loading, the composite retains that lower peak rate; the factors widen to 10.0 and 11.1, respectively.

The composite peak at 241.3720 °C lies below both precursors, 5.4 °C below alginate and 63.8 °C below chitosan, not between them. Its peak position and absence of a 305 °C chitosan-like event are consistent with an alginate-associated decomposition pathway; the trace alone cannot prove the chemical mechanism.

After loading, the peak-rate maximum shifts down 4.40 °C from 241.3720 to 236.9691 °C. This is a lower peak temperature, not by itself proof of an earlier degradation onset. Sorbed phosphate and counter-ions may alter the alginate-rich network, but that mechanism is a hypothesis. Meanwhile the peak rate falls from 0.1187 to 0.1023 %/°C: peak temperature and peak rate are separate observations, neither establishing thermal stabilisation by itself.

For the recovered material, 5% of the mass remaining at 150 °C has been lost by approximately 195 °C under the nitrogen ramp. This defines the screening handling/regeneration ceiling, about 42 °C below its peak-rate maximum and roughly 20 °C above the benchmark's assumed 175 °C regeneration temperature. It is not a demonstrated safe service limit or a measurement of repeated-cycle capacity retention.